

Aditya Kulkarni

adityakulkarnius@gmail.com | linkedin.com/in/aditya-kulkarni-355b81217 | kulkarniaditya.com | github.com/Aditya-k24 |

Raleigh, NC (open to relocation)

PROFESSIONAL SUMMARY

Backend engineer specializing in agentic workflow orchestration and AI-native systems. Built durable multi-step workflows with Temporal.io at Isomer AI -- claims automation platform with 3 LLM providers (Claude, GPT-4o, Gemini) routed via MCP tooling across 394 commits and 7 production deployments. Designed event-driven AI orchestration reaching 2,000 async inference requests/second with Temporal + Kafka (AlgoPulse). Built a vendor-neutral LLM eval harness measuring hallucination rate, p50/p95 latency, and cost/call across 3 models. MS CS at NC State (May 2026). IEEE-published. Portfolio: kulkarniaditya.com

CORE COMPETENCIES

Python / FastAPI

Temporal.io Workflow Orchestration

Kafka

LLM Pipelines (Claude, GPT-4o, Gemini)

MCP Tooling

Kubernetes

AWS / GCP

PostgreSQL / Redis

Observability (Prometheus, Sentry, OTel)

TypeScript / Node.js

Java / Spring Boot

Docker / CI/CD

Distributed Systems

RAG / Vector Databases

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built Temporal.io workflow orchestration for AI-powered claims automation -- durable multi-step state machines handling claim lifecycle with fault-tolerant execution and real-time event pipelines (Pusher); 394 commits across 7 production deployments
- Integrated multi-model LLM routing across 3 providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models via MCP tooling; implemented RBAC, document processing pipelines (PDF, email via Microsoft Graph API, S3), and Auth0 authentication in TypeScript, Node.js, Express, PostgreSQL, Redis, and Prisma

NC State University April 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform (JBrowse 2 on Google Cloud Run) serving 500GB of strawberry and grape datasets to 100+ researchers worldwide
- Applied hyperparameter tuning and time-series analysis to ML architecture, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated Grafana monitoring dashboards; supported model red-teaming for reliability

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed data pipelines with Hadoop + Spark for batch processing and Kafka streaming for real-time ingestion; reduced ticket-processing latency by 20% and processing time by 15%

PROJECTS

[AlgoPulse - Agentic AI Orchestration Platform](#) Temporal + Kafka

Identified API-polling bottleneck under heavy AI inference load; profiled and re-architected to event-driven streaming. Supabase Edge Functions stream AI reasoning loops to the client in real time. Temporal.io for durable workflow state; Kafka for worker coordination and fault tolerance. Reached 2,000 async inference requests/second with zero message loss under load.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

[LLM Eval Harness](#) AI/ML Ops

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on a 2,000-example frozen test set (3 repeated runs). Tracked ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call per model. Statistical significance via paired bootstrap CIs, McNemar test, and Holm correction. JSONL + Parquet storage; DuckDB analytics; Streamlit dashboard.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

[VectorLift - Semantic Search Engine](#) Python / FastAPI

Production-grade semantic search on MS MARCO: BM25 baseline (Elasticsearch), dense bi-encoder retrieval (sentence-transformers, in-batch negatives), cross-encoder re-ranker. Kafka-driven indexing into Elasticsearch and Qdrant; FastAPI serving with per-stage latency tracking; Streamlit evaluation dashboard. NDCG@10/MRR@10/MAP with bootstrap significance testing.

Python, PyTorch, Elasticsearch, FAISS, Qdrant, Kafka, FastAPI, Prometheus, Grafana, Docker

[Glance AI - Privacy-First RAG System](#) Compliance RAG

Per-tenant data isolation architecture at the vector retrieval layer for fintech/healthcare environments. Each retrieval context is scoped to its tenant boundary before LLM input -- prevents cross-tenant contamination. Deterministic output patterns for regulated-data hallucination mitigation.

Python, LLMs, Vector Databases, RAG

[PulseRank - Real-Time Leaderboard](#) Distributed Systems

Multi-store event-driven platform: MySQL outbox + idempotency keys, MongoDB TTL projections, Redis sorted sets for sub-ms leaderboard reads. Multi-store consistency validated with Testcontainers; hot paths benchmarked with k6; OpenTelemetry + Prometheus observability; GitHub Actions CI/CD.

TypeScript, NestJS, MySQL, MongoDB, Redis, OpenTelemetry, Prometheus, Docker

[Mapify - NLP Tool](#) IEEE Published

T5-transformer pipeline converting 1,500+ PDFs and images into mind maps and flowcharts. Reduced academic review time by 60%. Presented to 500+ attendees at IEEE conference.

Python, T5-transformer, NLTK, NetworkX

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms

Mumbai University -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Big Data Infra, Machine Learning, Data Warehousing, DS & Algorithms, Blockchain

AWARDS & LEADERSHIP

HackNC24 Top 10 -- GenAI Finance App (PerCent); top 10 of 100+ teams, 48-hour build 2024

IEEE Publication -- Mapify: T5-transformer PDF-to-mind-map; 500+ attendees 2022

Sanjeevani Health App -- Top 1% of 1,100+ submissions; 87% ML classification accuracy 2023

Google Developer Student Club Lead -- 4+ events, 500+ students 2022-2024

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, R, C

AI/ML: LangChain, MCP, Hugging Face, PyTorch, TensorFlow, RAG, OpenAI/Anthropic/Google SDKs

Backend: FastAPI, Node.js, Spring Boot, Express, NestJS, REST APIs **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant

Infrastructure: AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, Terraform, Spark, Hadoop

Observability: Prometheus, Grafana, Sentry, OpenTelemetry **DevOps:** GitHub Actions, Docker Compose, Testcontainers, Flyway